

Coding & Solution

Date	03 July 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	[Paste your GitHub / GitLab or hosted link here]
Programming Language(s)	Python, HTML, CSS, JavaScript
Framework(s) Used	[Flask, Scikit-learn
Key Features Implemented	Crop Prediction, Input Validation, Data Preprocessing, Machine Learning Model, Prediction Result Display
Pending / Incomplete Features	None (or "Future work: Weather API Integration")
Setup / Run Instructions	Install requirements (pip install -r requirements.txt), run app.py, open http://127.0.0.1:5000 in browser

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	yes
2	Meaningful variable and function names are used	yes
3	Code includes comments / documentation where necessary	yes
4	Error handling is implemented for critical operations	yes
5	The application runs without critical errors	yes
6	Code is committed to a version control repository	yes

Additional Notes / Comments:

The OptiCrop project is developed using Flask and Scikit-learn with a modular architecture. The application performs crop prediction based on agricultural parameters using a trained machine learning model. Proper input validation, preprocessing, and error handling have been implemented to ensure reliable predictions. The code is organized, reusable, and easy to maintain for future enhancements such as weather API integration and fertilizer recommendation.